FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 16) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

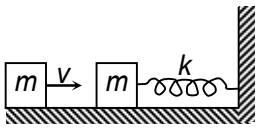
(Only One Option Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. Which of the following is the most accurate instrument for measuring length?
   (A) vernier calipers having 20 divisions on the sliding scale which coincide with 19 divisions on the main millimeter scale
   (B) a screw gauge having pitch 1 mm and 50 divisions on the circular scale
   (C) a vernier scale of least count 0.01 mm
   (D) a screw gauge of least count 0.001 mm

2. A ball is thrown from the ground to clear a wall 3 m high at a distance of 6 m and falls 18 m away from the wall, the angle of projection of ball is
   (A) $\tan^{-1}\left(\frac{3}{2}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$
   (C) $\tan^{-1}\left(\frac{1}{2}\right)$ (D) $\tan^{-1}\left(\frac{3}{4}\right)$

3. A block of mass m moving with velocity v strikes elastically another identical mass connected to a spring as shown in figure. The maximum compression produced in the spring (Assume surfaces to be smooth) is



   (A) $\sqrt{\frac{mv^2}{2k}}$ (B) $\sqrt{\frac{mv^2}{k}}$
   (C) $\sqrt{\frac{mv^2}{3k}}$ (D) none of these

Space for Rough work

4. A cube of side l and mass M is placed on rough horizontal surface and the friction is sufficient so that it will not move, if a constant force F = Mg is applied horizontally l/4 above the surface. Then the torque due to normal force about center of the cube is equal to

(A) $\frac{Mgl}{2}$ (B) $\frac{Mgl}{4}$

(C) $\frac{Mgl}{8}$ (D) zero

5. The work done to take a particle of mass m from surface of the earth to a height equal to 2R is (R is radius of earth)

(A) 2 mgR (B) $\frac{mgR}{2}$

(C) 3 mgR (D) $\frac{2mgR}{3}$

6. A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi}$ N-s/m^2, then velocity of ball will become 0.5 m/s after a time

(A) ln 4 s (B) 2 ln 4 s

(C) 3 ln 4 s (D) 2 ln 2 s

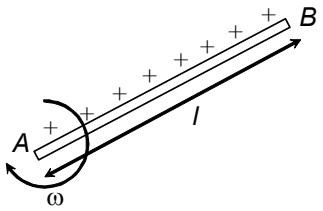
7. Two straight long conductors AOB and COD are perpendicular to each other and carry currents i_1 and i_2. The magnitude of the magnetic induction at a point P at a distance a from the point O in a direction perpendicular to the plane ACBD is

(A) $\frac{\mu_0}{2\pi a}(i_1 + i_2)$ (B) $\frac{\mu_0}{2\pi a}(i_1 - i_2)$

(C) $\frac{\mu_0}{2\pi a}\left(i_1^2 + i_2^2\right)^{1/2}$ (D) $\frac{\mu_0}{2\pi a}\frac{i_1 i_2}{(i_1 + i_2)}$

Space for Rough work

8. A non-conducting rod AB of length l has a positive linear charge density λ. The rod is rotated about point A with an angular velocity ω in the plane of paper. The magnetic moment of the rod is



(A) $\frac{\lambda\omega l^3}{2}$ (B) $\frac{2\lambda\omega l^3}{3}$

(C) $\frac{3\lambda\omega l^3}{2}$ (D) $\frac{\lambda\omega l^3}{6}$

9. A bird is flying over a swimming pool at a height of 2m from the water surface. If the bottom is perfectly plane reflecting surface and depth of swimming pool is 1 m, then the distance of final image of bird from the bird itself is ($\mu_w = 4/3$)

(A) $\frac{11}{3}$m (B) $\frac{23}{3}$m

(C) $\frac{11}{4}$m (D) $\frac{11}{2}$m

10. The mean lives of a radioactive sample are 30 years and 60 years for α-emission and β-emission respectively. If the sample decays both by α-emission and β-emission simultaneously, the time after which, only one-fourth of the sample remain is

(A) 10 years (B) 20 years

(C) 40 years (D) 45 years

Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 11 & 12

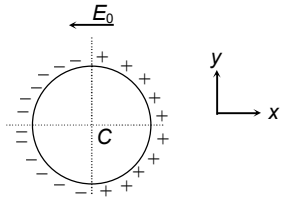
The position vector of a body of mass m = 4 kg is given as $\vec{r} = \hat{i}(t^2 - 4t) + \hat{j}(-t^3)$, where $\vec{r}$ is in metres and t in seconds.

11. The magnitude of torque with respect to origin, acting on the particle at t = 1s will be
(A) zero (B) 48 Nm
(C) 64 Nm (D) 80 Nm

12. Work done by force acting on it in first two seconds will be
(A) 32 J (B) 256 J
(C) 320 J (D) 288 J

Paragraph for Questions 13 & 14

A uniform ring of mass m and radius R can rotate freely about an axis passing through centre C and perpendicular to plane of paper. Half of ring is positively charge and other half is negatively charge. Uniform electric field E_0 is switched on along –ve x-axis (Axis are shown in figure)
[magnitude of charge density λ]



13. The dipole moment of ring is
(A) $2\lambda R^2$ (B) $4\lambda R^2$
(C) $2\pi\lambda R^2$ (D) $4\pi\lambda R^2$

14. If ring is slightly disturb from given position, find the angular speed of ring when it rotate by $\pi/2$.
(A) $2\sqrt{\frac{\lambda E_0}{m}}$ (B) $\sqrt{\frac{\lambda E_0}{m}}$
(C) $\sqrt{\frac{8\lambda E_0}{m}}$ (D) none

Space for Rough work

Paragraph for Questions 15 & 16

An ammeter and a voltmeter are connected in series to a battery with emf E = 6 volt and negligible resistance. When a resistance R = 3Ω is connected in parallel to voltmeter, reading of ammeter increases three times while that of voltmeter reduces to one third.

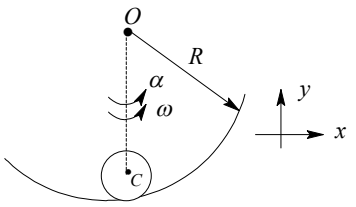
15. Reading of voltmeter after the connection of resistance is
(A) 1 Volt (B) 3 Volt
(C) 9/2 Volt (D) 3/2 Volt

16. Reading of ammeter before the connection of the resistance is
(A) $\frac{3}{4}$A (B) $\frac{6}{7}$A
(C) $\frac{3}{16}$A (D) 1 A

(Matching List Type)

This section contains **4 multiple choice questions. Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Line joining the centre of path and cylinder is having angular velocity ω and angular acceleration α at the given instant: Match the statements from List I with those in List II and select the correct answer using the code given below the lists.



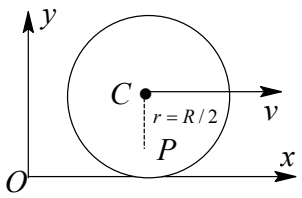
	List-I		List-II
(P)	Rotational component of kinetic energy of cylinder	1.	$(R-r)\alpha\hat{i}+(R-r)\omega^2\hat{j}$
(Q)	Kinetic energy of cylinder	2.	$\frac{\omega^2R(R-r)}{r}\hat{j}$
(R)	Acceleration of centre of cylinder	3.	$\frac{1}{4}m(R-r)^2\omega^2$
(S)	Acceleration of point of contact	4.	$\frac{3}{4}m(R-r)^2\omega^2$

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	4	1	3
(C)	1	2	3	4
(D)	4	1	2	3

Space for Rough work

18. A disc of mass m and radius R is rolling without slipping on a horizontal fixed rough surface with tramlational velocity v as shown in the figure. Here $\vec{L}_o, \vec{L}_p$ and k_o, k_p represents the angular momentum and kinetic energy about point O and P respectively: Match the physical quantities from List I with those in List II and select the correct answer using the code given below the lists.

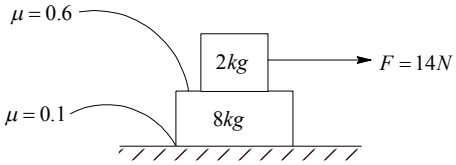


	List-I		List-II
(P)	$\vec{L}_o$	1.	$\frac{3}{4}mVR(-R)$
(Q)	$\vec{L}_p$	2.	$\frac{3}{4}mVR^2$
(R)	K_o	3.	$\frac{3}{2}mVR(-\hat{k})$
(S)	K_p	4.	$\frac{3}{8}mv^2$

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	4	1	3
(C)	3	1	2	4
(D)	4	1	2	3

19. A block of mass 2kg is kept over a block of mass 8 kg as shown in the figure. A force $F = 14N$ acts on the upper block. Match the physical quantities from List I with those in List II and select the correct answer using the code given below the lists. All values are in S.I. units:



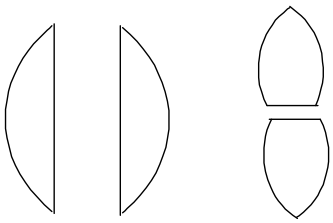
	List-I		List-II
(P)	Acceleration of block 2 kg	1.	10
(Q)	Acceleration of block 8 kg	2.	1
(R)	Friction action between the blocks	3.	12
(S)	Friction acting between lower block and ground	4.	$\frac{1}{4}$

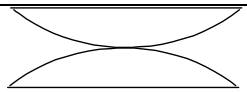
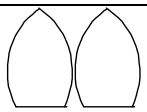
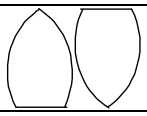
Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	4	1	3
(C)	1	2	3	4
(D)	2	4	3	1

Space for Rough work

20. A thin biconvex lens of small aperture and having focal length 30 cm is cut tow ways as shown in the figure. The focal length of different combination given in List-I with their value in List-II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.



	List-I		**List-II**
(P)		1.	Infinite
(Q)		2.	30 cm
(R)		3.	60 cm
(S)		4.	15 cm

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	3	4	1
(C)	1	2	3	4
(D)	2	4	3	1

Space for Rough work

Chemistry

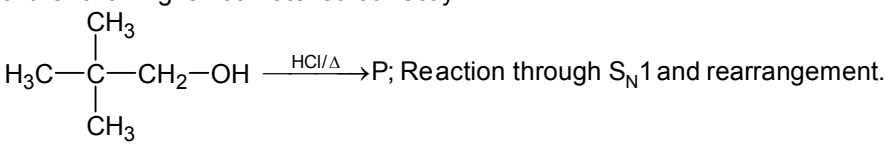
PART – II

SECTION – A

(Only One Option Correct Type)

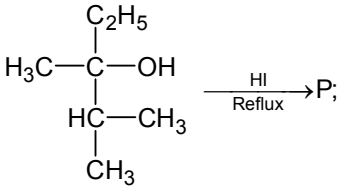
This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

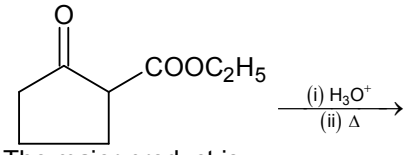
1. Which of the following is not matched correctly?

(A) 

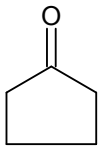
(B) $CH_3-CH_2-OH \xrightarrow{SOCl_2} P$; Reaction through S_Ni.

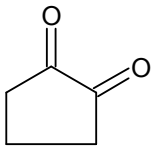
(C) $CH_3-CH_2-OH \xrightarrow[Reflux]{HI} P$; Reaction through S_N2 reaction.

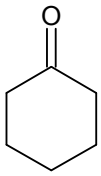
(D) 

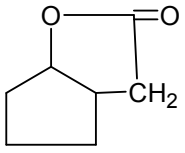
2. 

The major product is

(A) 

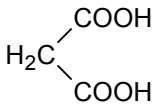
(B) 

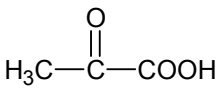
(C) 

(D) 

Space for Rough work

3. $CH_3-COOH \xrightarrow[Red\ P]{Br_2(1eq)} (A) \xrightarrow{KCN} (B) \xrightarrow{H_2O/H^+} (C) \xrightarrow{\Delta} (D)$, Product D is

(A) 

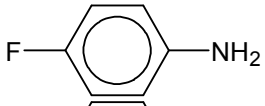
(B) 

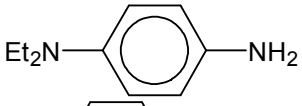
(C) CH_3COOH

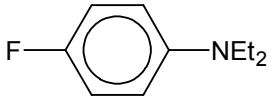
(D) CH_4

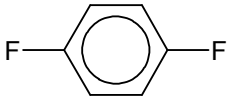
4. 

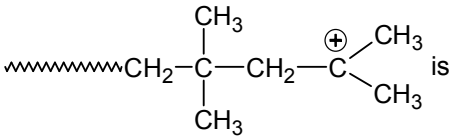
What is 'Z'?

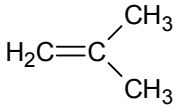
(A) 

(B) 

(C) 

(D) 

5. The monomer of the polymer:  is

(A) 

(B) $(H_3C)_2C{=}C(CH_3)_2$

(C) $H_3CHC{=}CHCH_3$

(D) $H_3C{-}CH{=}CH_2$

6. State of hybridization of sulphur, carbon-1 and carbon-2 in $F_3S\overset{1}{C}\overset{2}{C}F_3$ respectively are"

(A) sp^3, sp^3, sp^3
(B) sp^3, sp^2, sp^3
(C) sp^3d, sp, sp^3
(D) sp^3, sp, sp^3

Space for Rough work

7. A XF_4 type molecule have $\mu = 0$, which additional information is required to conform the geometry of molecule
(A) All the X—F bond length are identical
(B) Molecule has same F—X—F bond angle between any two adjacent F
(C) Number of lone pair of electron on central atom ≤ 2
(D) Planarity of molecule

8. Solid AB haz ZnS type structure. If the radius of A^+ ion is 22.5 pm then radius of B^- ion will be:
(A) 100 pm (B) 200 pm
(C) 150 pm (D) 95 pm

9. $2Zn + O_2 \longrightarrow 2ZnO;\ \Delta G^o = -616$ J

$2Zn + S \longrightarrow 2ZnS;\ \Delta G^o = -293$ J

$2S + 2O_2 \longrightarrow 2SO_2(g);\ \Delta G^o = -408$ J

ΔG^o for the following reaction:

$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$, would be:

(A) –731 J (B) –1317 J
(C) +731 J (D) +1317 J

10. A gas shows heating effect on sudden expansion. It shows that:
(A) the gas is nobel
(B) the gas is ideal
(C) the inversion temperature of the gas is very low
(D) its Vanders Waal's constant 'b' is high

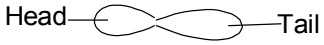
Space for Rough work

Comprehension Type (Only One Option Correct)

This section contains **3 paragraphs** each describing theory, experiment, data etc. **Six questions** relate to four paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 11 to 12

Hybridisation is a process of mixing of atomic orbitals to give mixed or hybrid orbitals. Hybrid orbitals have equal energy and their hypothetical shape may be given as



Five d-orbitals are non-degenerate, they are divided into two different set of orbitals

$e_g \rightarrow d_{x^2-y^2}, d_{z^2}$

$t_{2g} \rightarrow d_{xy}, d_{y^2}, d_{zx}$

Hybridisation involving d-orbitals are dsp^2, sp^3d, dsp^3, sp^3d^2, d^2sp^3, sp^3d^3

11. Which of the following orbitals are involved in sp^3d^2 hybridisation?
(A) d_{xy}, d_{yz} (B) $d_{x^2-y^2}, d_{xy}$
(C) $d_{x^2-y^2}$, d_{z^2} (D) d_{z^2}, d_{xy}

12. Which of the following d – orbitals are involved in sp^3d^3 hybridisation?
(A) $d_{x^2-y^2}, d_{z^2}, d_{xy}$ (B) d_{xy}, d_{yz}, d_{zx}
(C) $d_{x^2-y^2}, d_{xy}, d_{xz}$ (D) d_{z^2}, d_{y^2}, d_{zx}

Space for Rough work

Paragraph for Question Nos. 13 to 14

If a cell has cell potential 'E' and standard cell potential 'E^o', then free energy change of cell process may be calculated as,

$\Delta G = -W = -nFE$ and $\Delta G^o = -W_{max} = -nFE^o$

Where 'n' is the number of electrons involved in overall all process . According to Gibs-Helmholtz equation:

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = \Delta H + T\left(\frac{d\Delta G}{dT}\right)_P$$

Temperature coefficient of cell 'μ' will be equal to $\left(\frac{dE}{dT}\right)_P$

13. ΔG^o for the Daniell cell

$Zn(s)|ZnSO_4||CuSO_4|Cu(s)$

$E^o_{Zn^{2+}/Zn} = -0.76V$; $E^o_{Cu^{2+}/Cu} = +0.34$ V

Will be:

(A) –312.3 KJ (B) –212.3 KJ

(C) –123.2 KJ (D) –323.1 KJ

14. The temperature coefficient of a cell whose cell reaction is

$$Pb(s) + HgCl_2(aq) \longrightarrow PbCl_2(aq) + Hg(\ell)$$

$\left(\frac{dE}{dT}\right)_P = 1.5 \times 10^{-4}$ vK^{-1} at 298 K

The change in entropy in JK^{-1} mol^{-1} for the given cell reaction will be

(A) 14.475 (B) 57.9

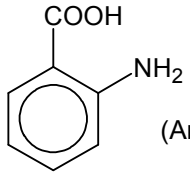
(C) 28.95 (D) 86.82

Space for Rough work

Paragraph for Question Nos. 15 to 16

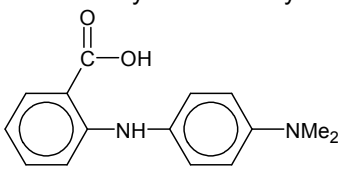
Methyl red is commonly used as indicator for acid base titrations. It is prepared by treating $NaNO_2$/HCl with anthranilic acid and the resulting solution is mixed with N, N-dimethyl aniline and shaked well the solution for some minute to get 'Methyl Red".

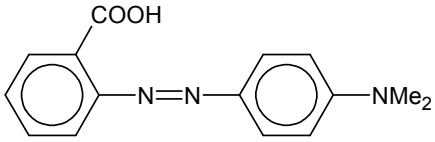
Given

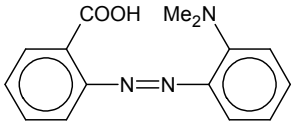


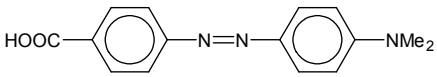
(Anthranilic Acid)

15. Which is most likely to be "Methyl Red"?

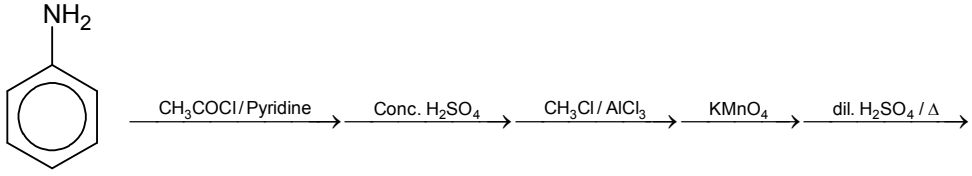
(A) 

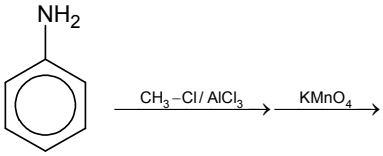
(B) 

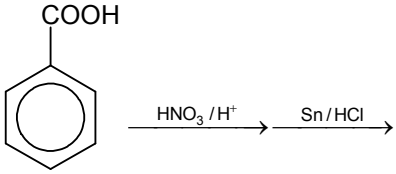
(C) 

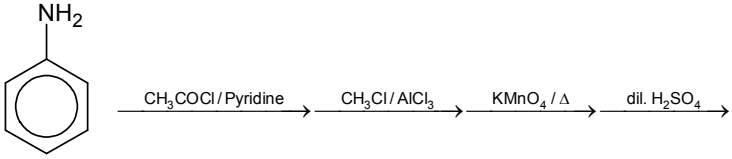
(D) 

16. Which method can be prepare o-amino benzoic acid, the main raw material?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

(Matching List Type)

This section contains **4 multiple choice questions. Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

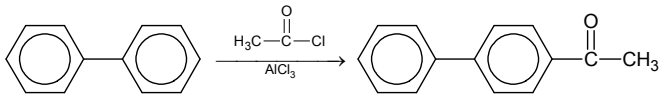
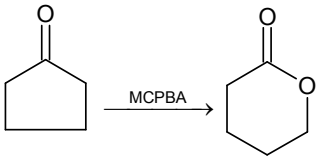
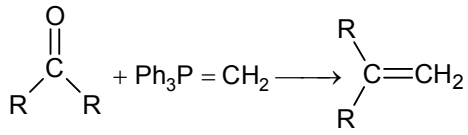
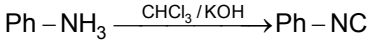
17. Match the list- I with list – Ii

	List– I		**List – II**
(P)	Calomel	(1)	Reference
(Q)	Glass	(2)	Redox
(R)	Hydrogen	(3)	Membrane
(S)	Quinohydrone	(4)	Gas

Codes:

	P	Q	R	S
(A)	1	3	4	2
(B)	3	1	2	4
(C)	1	2	4	3
(D)	1	4	2	3

18. Match List – I with List – II.

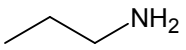
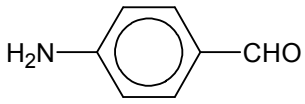
	List – I Transformation		**List – II Name**
(P)		(1)	Hofmann Isocyanide
(Q)		(2)	Wittin Reaction
(R)		(3)	Bayer-Villiger oxidation
(S)		(4)	Friedel Craft's Acylation

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	1	4	2
(C)	1	2	3	4
(D)	1	3	2	4

Space for Rough work

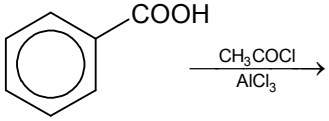
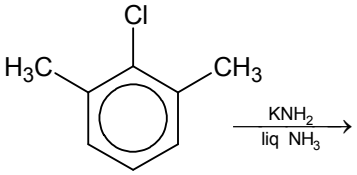
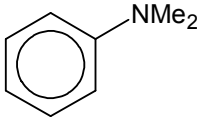
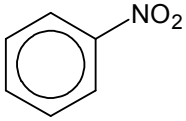
19. Match the list- I with list – II

List– I		**List – II**
(P)	$H_3C-C\equiv CH$	(1) Positive test with Fehling's solution
(Q)	HCOOH	(2) Positive test with Tollen's reagent
(R)		(3) Decolourise Br_2 water
(S)		(4) Isocyanide test

Codes:

	P	Q	R	S
(A)	2,3	1,2	4	2,4
(B)	1, 2, 3	1,2	2	4
(C)	2	1, 2	1, 2, 3, 4	4
(D)	1	4	2	3

20. Match the list- I with list – II

List– I		**List – II**
(P)		(1) O, P – directing compound
(Q)		(2) Activated compound
(R)		(3) No reaction
(S)		(4) Deactivated compound

Codes:

	P	Q	R	S
(A)	2	1, 2	3, 4	4
(B)	4	1, 2, 3	1, 2	4
(C)	1, 2	3, 4	1, 2	2, 3
(D)	1	2, 3	2	1, 4

Space for Rough work

Mathematics PART – III

SECTION – A

(Only One Option Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. $\tan^{-1}(\sin x) = \sin^{-1}(\tan x)$ holds true for

   (A) $x \in R$ (B) $2n\pi - \frac{\pi}{2} \le x \le 2n\pi + \frac{\pi}{2}$; $(n \in z)$

   (C) $x \in n\pi$, $(n \in z)$ (D) $x \in \{0, z^+\}$

2. Let P(x) be a polynomial with degree 2009 and leading co-efficient unity such that P(0) = 2008, P(1) = 2007, P(2) = 2006, ….. P(2008) = 0 and the value of P(2009) = $(\underline{|n}) - a$ where n and a are natural number than value of (n + a) is

   (A) 2008 (B) 2009

   (C) 2010 (D) 2011

3. Let $f(x) = x^3 + x^2 + 100x + 7 \sin x$, then equation $\frac{1}{y - f(1)} + \frac{2}{y - f(2)} + \frac{3}{y - f(3)} = 0$ has

   (A) no real root (B) one real root

   (C) two real roots (D) more than two real roots

4. Let $(1+x)^n = \sum_{r=0}^{n} C_r x^r$ and $u_n = \sum_{r=0}^{n} (-1)^r \frac{C_r}{x+r}$ at x = 2. Then the sum to infinity of $u_1 + u_2 + u_3 + \ldots$ is

   (A) 0 (B) $\frac{1}{2}$

   (C) 1 (D) 2

Space for rough work

5. Consider the system of equations ax + by = 0 and cx + dy = 0 where a, b, c, d $\in$ {1, 2}. The probability that the system of equations has a unique solution is

(A) $\frac{3}{8}$ (B) $\frac{5}{16}$

(C) $\frac{9}{16}$ (D) $\frac{5}{8}$

6. If circumcentre of an equilateral triangle inscribed in $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with vertices having eccentric angles α, β, γ respectively is (x_1, y_1) then $\sum\cos\alpha\cos\beta+\sum\sin\alpha\sin\beta$ is

(A) $\frac{9x_1^2}{2a^2}+\frac{9y_1^2}{2b^2}-\frac{3}{2}$ (B) $\frac{x_1^2}{2a^2}+\frac{y_1^2}{2b^2}-\frac{5}{2}$

(C) $\frac{x_1^2}{9a^2}+\frac{y_1^2}{9b^2}-\frac{5}{9}$ (D) $\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-\frac{1}{2}$

7. In a ΔABC, CD is the bisector of the angle C. If $\cos\frac{C}{2}$ has the value $\frac{1}{3}$ and CD = 6, then $\frac{ab}{a+b}$ is equal to

(A) 3 (B) 6

(C) 9 (D) 18

8. Let $S(x)=\int\frac{dx}{e^x+8e^{-x}+4e^{-3x}}$, $R(x)=\int\frac{dx}{e^{3x}+8e^x+4e^{-x}}$ and M(x) = S(x) – 2R(x).

If $M(x)=\frac{1}{2}\tan^{-1}(f(x))+c$ where c is an arbitrary constant then $f(\log_e 2)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Space for rough work

9. f(x) is a differentiable function satisfying the relation $f(x) = x^2 + \int_0^x e^{-t} f(x-t)\,dt$, then $\sum_{k=1}^{9} f(k)$ is equal to

(A) 960 (B) 1060
(C) 1224 (D) 1260

10. The solution of differential equation $x^2(x\,dy + y\,dx) = (xy - 1)^2\,dx$ is (where c is an arbitrary constant)

(A) $xy - 1 = cx$ (B) $xy - 1 = cx^2$
(C) $\frac{1}{xy-1} = \frac{1}{x} + c$ (D) none of these

Comprehension type (Only One Option Correct)

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:
Let each of the circles
$S_1 = x^2 + y^2 + 4y - 1 = 0$
$S_2 = x^2 + y^2 + 6x + y + 8 = 0$
$S_3 = x^2 + y^2 - 4x - 4y - 37 = 0$
touches the other two. Let P_1, P_2, P_3 be the point of contact of S_1 and S_2, S_2 and S_3, S_3 and S_1 respectively. Let T be the point of concurrence of the tangents at P_1, P_2, P_3 to the circles. C_1, C_2, C_3 are the centres of S_1, S_2, S_3 respectively

11. P_2 and P_3 are reflections of each other in the line

(A) $y = x$ (B) $y = x + 1$
(C) $2x - y + 3 = 0$ (D) $2x + y + 9 = 0$

12. The area of the quadrilateral $TP_2C_3P_3$ is

(A) 11 sq. units (B) 25 sq. units
(C) 15 sq. units (D) 9 sq. units

Space for rough work

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If f(x) is a differentiable function wherever it is continuous and $f'(c_1) = f'(c_2) = 0$, $f''(c_1)f''(c_2) < 0$, $f(c_1) = 5$, $f(c_2) = 0$ and $(c_1 < c_2)$

13. If f(x) is continuous in $[c_1, c_2]$ and $f''(c_1) - f''(c_2) > 0$, then minimum number of roots of $f'(x) = 0$ in $[c_1 - 1, c_2 + 1]$ is
(A) 2 (B) 3
(C) 4 (D) 5

14. If f(x) is continuous in $[c_1, c_2]$ and $f''(c_1) - f''(c_2) > 0$, then minimum number of roots of $f(x) = 0$ in $[c_1 - 1, c_2 + 1]$ is
(A) 2 (B) 3
(C) 4 (D) 5

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:
Define a function ϕ: N $\rightarrow$ N as follows: $\phi(1) = 1, \phi(p^n) = p^{n-1}(p-1)$ if p is prime and n $\in$ N and $\phi(mn) = \phi(m)\,\phi(n)$, if m and n are relatively prime natural numbers then

15. $\phi(8n + 4)$ where n $\in$ N is equal to
(A) $2\phi(4n + 2)$ (B) $\phi(2n + 1)$
(C) $2\phi(2n + 1)$ (D) $4\phi(2n + 1)$

16. The number of natural numbers 'n' such that $\phi(n)$ is odd is
(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

(Match List Type)

This section contains 4 multiple choice questions. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

17. Match the following List–I with List–II

List – I		List – II
(P)	Let $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = x^2 + xf'(2) + f''(3)$, then $f'(1) - f'(2)$ is equal to	1. 2
(Q)	If $f(x - y)$, $f(x)\cdot f(y)$ and $f(x + y)$ are in A.P., for all x, y and $f(0) \neq 0$, then $f'(2) + f'(-2)$ is equal to	2. 1
(R)	If $f(x) = x^3 + x^2\cdot f'(1) + xf''(2) + f'''(3)$ for all x, then $f(0) + f(3) + 1$ is equal to	3. 4
(S)	Let $f(x) = x^n$, n being a positive integer, then value of n for which the equality $f'(a + b) = f'(a) + f'(b)$ is valid for all a, b > 0 is	4. 0

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	2	1
(C)	3	4	2	1
(D)	1	2	3	4

18. A function F is defined by $F(x) = \int_1^x \frac{e^t}{t}dt \ \forall\ x > 0$. Now express the functions in List–I in terms of F, then match the following List–I with List–II

List – I	List – II
(P) $\int_1^x \frac{e^t}{t+2}dt$	1. $F(x) - \frac{e^x}{x} + e$
(Q) $\int_1^x \frac{e^{3t}}{t}dt$	2. $xe^{\frac{1}{x}} - e - F\left(\frac{1}{x}\right)$
(R) $\int_1^x \frac{e^t}{t^2}dt$	3. $e^{-2}[F(x + 2) - F(3)]$
(S) $\int_1^x e^{\frac{1}{t}}dt$	4. $F(3x) - F(3)$

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	1	2
(C)	3	4	2	1
(D)	1	2	3	4

Space for rough work

19. Match the following List–I with List–II

List – I		List – II	
(P)	An urn contains five balls, two balls are drawn and are found to be white. If probability of all the balls in urn are white is k, then	1.	$k = \frac{31}{91}$
(Q)	Out of 15 consecutive integers three are selected at random, then the probability of the sum is divisible by 3 is k, then	2.	$k = \frac{3}{16}$
(R)	If 3 cards are placed at random and independently in 4 boxes lying in a straight line. Then the probability of the cards going into 3 adjacent boxes is k, then	3.	$k = \frac{7}{8}$
(S)	A box contains 4 balls which are either red or black, 2 balls are drawn and found to be red if these are replaced, then the probability that next draw will result in a red ball is k, then	4.	$k = \frac{1}{2}$

Codes:

	P	Q	R	S
(A)	4	1	3	2
(B)	1	2	3	4
(C)	3	4	2	1
(D)	4	1	2	3

20. Match the following List–I with List–II

List – I		List – II	
(P)	$\sin\left[\frac{\pi}{900}\left\{\sum_{r=1}^{10}(r-\omega)\left(r-\omega^2\right)\right\}\right]$ is equal to	1.	0
(Q)	if root of $t^2 + t + 1 = 0$ be α, β, then $\alpha^4 + \beta^4 + \alpha^{-1}\beta^{-1}$ is equal to	2.	4
(R)	if $\left[\frac{1+\cos\theta+i\sin\theta}{\sin\theta+i(1+\cos\theta)}\right]^4 = \cos n\theta + i\sin n\theta$, then n is equal to	3.	i
(S)	if $z_r = \cos\frac{\pi}{3^r} + i\sin\frac{\pi}{3^r}$ (where r = 1, 2, 3, ...), then value of $z_1z_2z_3$ ..... ∞ is equal to	4.	1

Codes:

	P	Q	R	S
(A)	4	1	3	2
(B)	1	4	2	3
(C)	2	3	4	1
(D)	4	1	2	3

Space for rough work